

Reflections on Socio-political, Cultural and Environmental Challenges through Computational Media Arts and Design

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ABSTRACT

The field of emerging computational media arts and design is rapidly evolving as an interdisciplinary and high-impact domain. Beyond its technological affordances, it could also serve as a lens for addressing the pressing environmental, social, cultural, and political challenges. Within this context, this panel highlights the contributions of four women that shed light on the nuanced topics of inclusivity, transparency, accessibility, awareness, and action. These experts bring diverse perspectives and engage with technical advancements critically through their practice and research.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts.

KEYWORDS

Computational Media, Ethics, Computer Vision, Interactive Media

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1 INTRODUCTION

Against the backdrop of growing environmental, social, cultural, and political challenges, emerging technologies and computational media arts can play a pivotal role in addressing these issues. It can serve as a powerful and critical tool to conduct interdisciplinary research, visualize hidden data patterns, provoke discussions with stakeholders, and ultimately contribute to ethical frameworks [Herruzo, 2022 2022].

State-of-the-art tools and techniques, such as machine, real-time graphics, data-driven visualizations, as well as physical and spatial

computing, when synergistically combined with critical design methodologies, open new avenues for addressing today's pressing issues. They present affordances to empower and enable new ways of addressing contemporary imperatives. Utilizing these tools with a critical and thoughtful approach allows for the promotion of inclusivity, the establishment of transparency, the advancement of accessibility [Boulamwini and Gebru, 2018], the cultivation of awareness, and the initiation of action.

2 DISCUSSION

This panel will focus on five key topics that are critical to the field of emerging computational media and the computer graphics community. Firstly, the topic of 'inclusivity' with a focus on inspiring the next generation of female (non-binary) leaders from minority communities working on computer graphics, spatial computing, and interactive technologies [Farahi, 2021 2021]. Secondly, the topic of 'accessibility' in the context of technology access and sharing. Thirdly, the topic of 'transparency,' based on data democracy and AI ethics, increases understanding and promotes accountability. Fourthly, the topic of 'awareness' emphasizes acknowledgment and taking responsibility for possible ethical incursions. Finally, the topic of 'action' highlights the importance of community building and cross-disciplinary collaborations for critical decision-making and fostering change [Williams, 2022 2021]. To create engaging and diverse outcomes, these topics will be addressed through the panelists' applied advanced technologies and research. These topics serve as a catalyst for change, sparking conversations and inspiring individuals to take action.

The panel will consist primarily of discussing the topics mentioned above between the moderator and the speakers designed to engage the audience during the Q&A session. This panel provides a unique opportunity for those interested in emerging computational media practices, such as computer graphics professionals, artists, educators, and industry representatives, to gain new insights into how their work could be used to address environmental, political, and social challenges, from some of the leading voices in the field. Figure 1 showcases authors' projects that highlight the transformative potential of emerging computational media and computer graphics.

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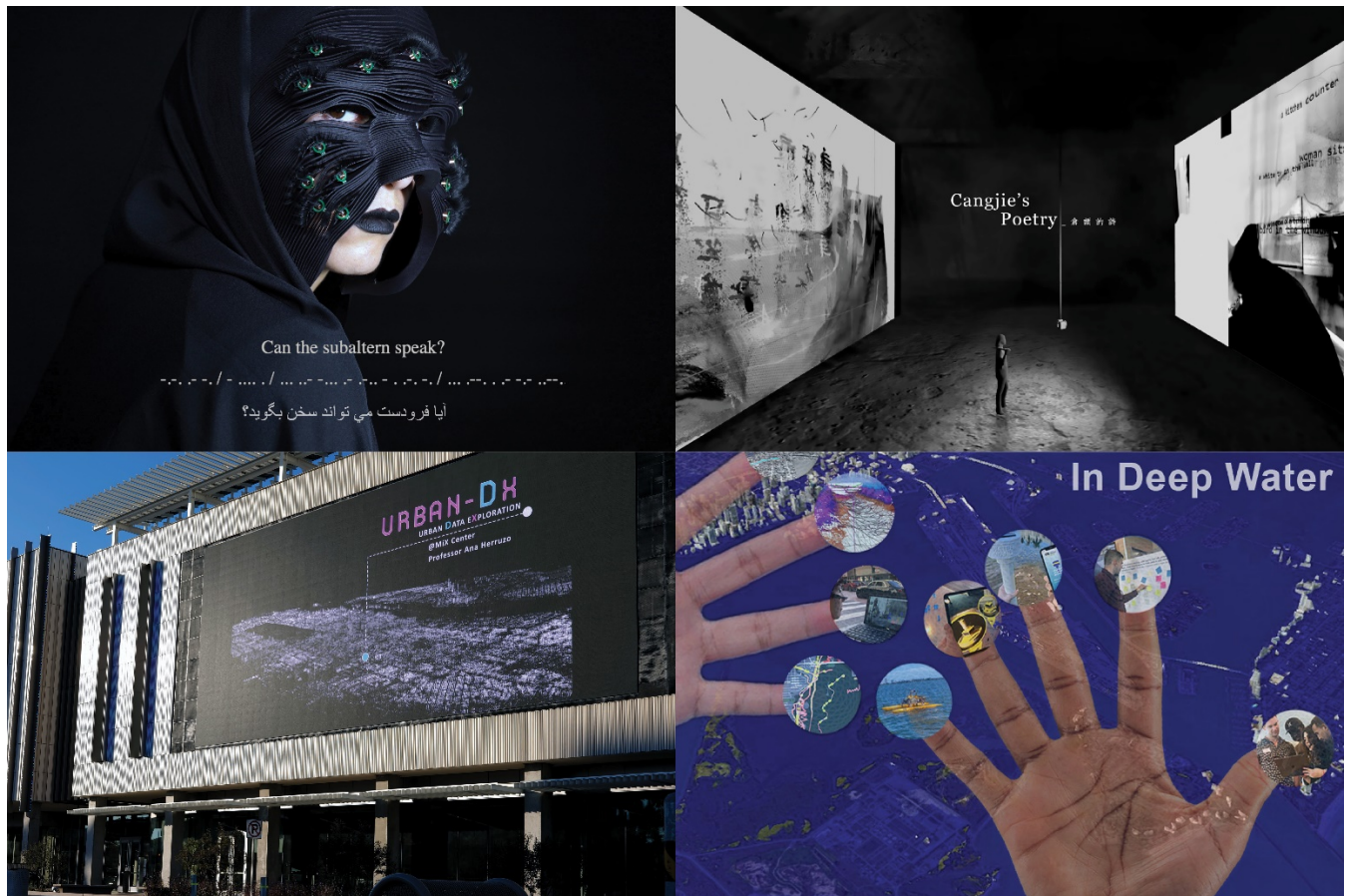


Figure 1: Top left: *Can the Subaltern Speak?* by Behnaz Farahi. Top Right: *Cangjie's Poetry* by Weidi Zhang; Bottom Left: *Mediated Membranes: Urban Data Explorations* by Ana Herruzo; Bottom Right: *In Deep Water* by Biayna Bogosian.

3 CONCLUSION

The development of a comprehensive and ethical framework for computational media arts and design remains a significant challenge. Without a suggestion of a perfect solution, in our panel discussion at the SIGGRAPH 2023 Conference, we aim to foster an open dialogue and delve into how the role of computational media art can be viewed as a call to action rather than mere technological advancement. By leveraging computational media as a critical tool, we can collectively work towards a more inclusive, just, and sustainable world addressing the complex ethical considerations that arise in the intersection of technology and creative expressions.

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